

S2 APPENDIX. List of species and their abundance, functional guild and mean biomass captured in all sampling points.

TABLE A. Number of individuals, by species, captured in all transects and functional guild and mean biomass of each species.

Species	Functiona l Guild	Mean Biomas s (mg)	Number of Individuals																				
			1400			1300			1200			1100			1000			900			800		
			Altitude (m a.s.l.)			Transects			I	I	II	I	I	II	I	I	II	I	I	II	I	I	II
			I	I	I	I	I	I	II	I	I	I	I	I	I	I	I	II	II	I	I	I	
<i>Agamopus unguicularis</i>	Paracoprid	3.8	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1
<i>Agamopus viridis</i>	Paracoprid	3.0	0	0	0	0	1	0	1	0	0	0	0	0	1	2	0	0	1	1	1	0	
<i>Anomiopus virescens</i>	Paracoprid	11.2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	
<i>Ateuchus vividus</i>	Paracoprid	21.7	1	0	0	0	0	0	0	0	0	1	0	1	15	6	8	1	0	0	0	0	0
<i>Canthidium aff barbacenicum</i>	Paracoprid	8.6	0	0	0	0	0	0	0	0	0	2	0	0	0	0	0	0	0	4	3	4	
<i>Canthidium sp 1</i>	Paracoprid	6.8					1		2	1	1	1									1	1	
<i>Canthidium sp 2</i>	Paracoprid	8.3	1	0	0	1	5	2	5	2	1	0	6	4	1	0	2	24	4	9	32	4	2
<i>Canthidium sp 3</i>	Paracoprid	11.2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	0	0	0
<i>Canthidium sp 4</i>	Paracoprid	8.2	0	0	0	0	0	0	0	0	0	1	0	5	0	0	1	0	1	0	0	0	0
<i>Canthidium sp 5</i>	Paracoprid	7.2		1																	1		
<i>Canthidium sp 6</i>	Paracoprid	11.8	2	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Canthidium sp 8</i>	Paracoprid	7.9	1	1	8	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0
			0	0	0	7	0	4	1	0	1	7	4	9	9	0	5	1	3	3	0	0	0

<i>Canthidium sp 10</i>	Paracoprid	14.7	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	17	8	3	21	0	2	
<i>Canthon aff podagricus</i>	Telecoprid	6.4	0	0	0	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0	0	0	
<i>Canthon lamproderes</i>	Telecoprid	72.9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10	9	2	
<i>Canthon sp 1</i>	Telecoprid	5.9	0	0	0	6	7	3	2	1	9	1	1	9	0	0	9	59	11	3	7	0	5	0
<i>Canthon sp 2</i>	Telecoprid	30.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	4	
<i>Canthon sp 3</i>	Telecoprid	18.4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	3	3	4	
<i>Canthon unicolor</i>	Telecoprid	45.3	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	4	1	
<i>Coprophanaeus cyanescens</i>	Paracoprid	690.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Coprophanaeus ensifer</i>	Paracoprid	2050.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	
<i>Coprophanaeus jasius</i>	Paracoprid	894.7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	
<i>Coprophanaeus milon magnoi</i>	Paracoprid	751.0	0	0	0	0	0	0	1	0	0	2	0	1	0	0	2	0	0	0	0	0	3	
<i>Deltochilum inaequale</i>	Telecoprid	582.3	5	3	0	0	0	6	6	0	4	0	0	0	0	0	1	0	0	0	0	0	0	
<i>Deltochilum komareki</i>	Telecoprid	74.4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	3	2	4	2	
<i>Deltochilum pseudoicarus</i>	Telecoprid	464.2	0	0	0	6	8	0	0	1	0	1	3	2	0	0	1	6	7	1	0	1	0	
<i>Dichotomius bos</i>	Paracoprid	347.7	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	3	3	
<i>Dichotomius crinicollis</i>	Paracoprid	385.2	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	1	0	4	
<i>Dichotomius depressicollis</i>	Paracoprid	338.3	0	1	0	2	7	3	9	0	5	8	7	8	20	0	3	49	41	3	33	6	3	
<i>Dichotomius glaucus</i>	Paracoprid	289.0	0	0	0	0	1	0	0	0	0	3	1	0	1	0	0	0	0	0	0	0	0	
<i>Dichotomius luctuosus</i>	Paracoprid	197.3	0	7	3	0	0	0	1	1	1	0	0	0	0	2	0	0	0	1	0	0	1	
<i>Dichotomius nisus</i>	Paracoprid	309.8	0	0	0	0	1	1	0	0	0	2	0	1	2	2	5	1	0	1	1	1	1	
<i>Digitonthophagus gazella</i>	Paracoprid	51.0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	

<i>Eurysternus nigrovirens</i>	Endocopri d	8.7	0	2	0	0	5	2	2	4	0	6	0	8	2	1	0	0	4	3	1	2	1	2
<i>Genieridium bidens</i>	Endocopri d	6.6	0	0	0	0	0	0	1	0	1	0	0	0	0	1	0	0	0	1	0	0	0	1
<i>Genieridium cryptops</i>	Endocopri d	6.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
<i>Homocopris sp 1</i>	Paracoprid	79.2	0	0	0	0	0	0	6	4	1	0	0	0	1	0	0	3	0	2	0	0	0	0
<i>Homocopris sp 2</i>	Paracoprid	40.4	0	0	0	0	0	0	0	7	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Homocopris sp 3</i>	Paracoprid	69.0	0	0	0	0	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Ontherus (Caelontherus) sp</i>	Paracoprid	26.0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Ontherus aff carinifrons</i>	Paracoprid	39.6	0	0	1	1	2	1	1	2	4	0	2	0	0	0	0	1	0	0	0	0	0	0
<i>Ontherus appendiculatus</i>	Paracoprid	58.2	0	0	0	0	0	0	0	0	0	0	0	7	2	1	0	1	1	2	2	2	2	
<i>Onthophagus aff buculus</i>	Paracoprid	11.1	2	1	2	0	0	0	1	0	0	0	0	1	1	0	0	1	0	1	2	0	0	
<i>Onthophagus aff hirculus</i>	Paracoprid	9.7	18 8	3 1	4 3	1 3	1 2	1 6	2 0		1 5		2 0	1 7		1 28	2 9			1 7		1 17	1 5	1 15
<i>Oxysternon palaemon</i>	Paracoprid	273.4																			13	2	4	
<i>Phanaeus dzidoi</i>	Paracoprid	130.0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1
<i>Phanaeus kirbyi ledezmai</i>	Paracoprid	202.0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
<i>Phanaeus palaeno</i>	Paracoprid	215.3	0	0	0	0	0	3	0	0	0	0	1	0	0	1	0	0	0	0	0	17	0	5
<i>Sulcophanaeus menelas</i>	Paracoprid	428.3		3	1			1						1										
			47	0	9	5	3	1	7	5	6	3	3	2	0	2	4	1	1	0	0	0	0	0
<i>Trichillum sp 1</i>	Endocopri d	0.7					2		1	3														
			0	0	0	1	7	0	0	3	5	1	2	1	1	1	5	31	18	8	4	3	9	
<i>Trichillum sp 2</i>	Endocopri d	0.9	0	1	0	0	0	0	0	0	0	0	0	0	0	4	0	0	2	0	0	0	0	0
						2	9		9	7	1	1	3											
<i>Uroxys sp 1</i>	Paracoprid	1.1	0	0	0	5	2	0	7	9	3	1	9	8	0	1	0	30	14	8	0	1	0	0

<i>Uroxys sp 2</i>	Paracoprid	0.6	0	2	2	2	0	0	4	4	3	1							1				
							1	2		4	3		3										
<i>Uroxys sp 3</i>	Paracoprid	1.5	0	0	0	6	7	0	9	2	3	4	2	3	0	0	0	24	0	6	9	0	0
<i>Uroxys sp 4</i>	Paracoprid	2.2	0	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	2	0	0
<i>Uroxys sp 5</i>	Paracoprid	12.0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure A presents the abundance ranking of all dung beetle communities recorded with their functional group highlighted. At all altitudes the first and second most abundant species together were at least three times more abundant than the third (at 1400 m a.s.l. they were 25 times!). In three of the seven altitude classes, most abundant species were at least twice more abundant than second. The most abundant functional group was at least twice more abundant than the next most abundant functional group in five of the altitude classes (at 1200 m a.s.l. it was 6 times). Small paracoprids were the dominant functional group at all altitudes, except for the lowest (800 m a.s.l.), where large paracoprids dominated (Fig A).

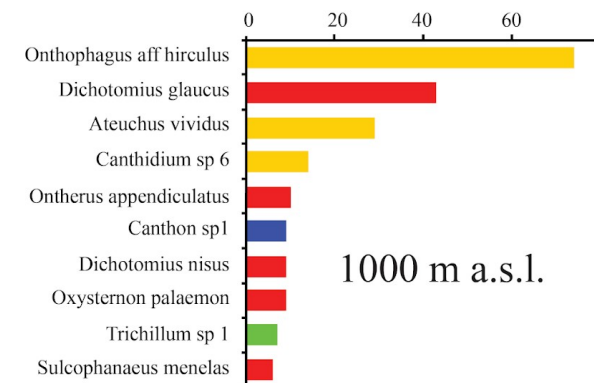
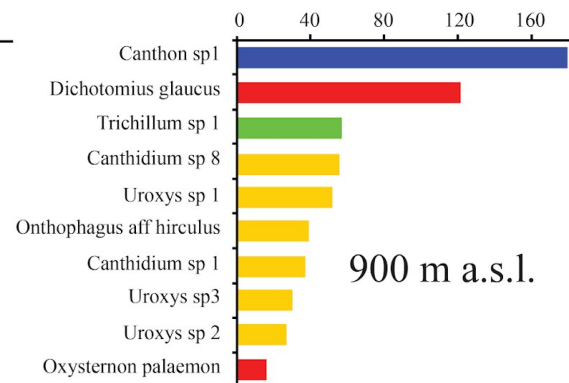
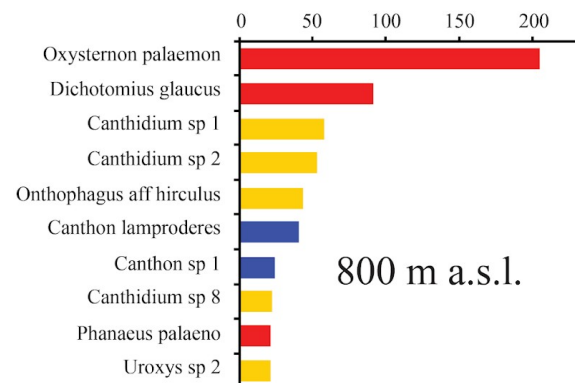


Fig A. Abundance rankings of dung beetle communities of all altitudes of the altitudinal gradient at Serra do Cipó, State of Minas Gerais, Brazil.
The colors highlight the functional group of each species.

